Capstone Lightning Talk

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Project Proposals & EDA

By: Zahraa Mohammed Alekri

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Three Dataset Proposals

01 — **US Accidents**
2016–2023 · 7.7M rows · 46 columns · Classification

02 — **Air Quality in India**
City-level AQI · Time series · Regression

03 — **Solar Power Generation**
Sensor data · Irradiance features · Regression

Dataset 01 · US Accidents (2016–2023)

Problem Statement

Traffic accidents across the US result in thousands of injuries and billions of dollars in emergency response costs annually. With no reliable way to anticipate accident severity at the time of occurrence, emergency services struggle to allocate resources efficiently. Using historical accident data enriched with weather, road, and location features, **can we predict the severity of a traffic accident (scale 1–4) to help traffic management authorities prioritize emergency response and reduce reaction time?**

Goal / Aim

Predict accident severity at the time of occurrence and **identify the strongest weather, road, and time-based risk factors.**

EDA Findings

Regression or Classification?

Classification
Target: **Severity (1–4 ordinal)**
Can simplify to binary:
Low (1–2) vs High (3–4)

Missing Values

End_Lat/Lng: **44%**
Precipitation: **28%**
Wind_Chill: **26%**
Wind_Speed: **7%**
Weather cols **~2%**

Categorical & Continuous Columns

13 Continuous (Temp, Humidity, Visibility…)
17 Categorical (State, Weather_Condition…)
13 Boolean (POI flags)

Outliers

Temperature: max **207°F**
Wind_Speed: max **1,087 mph**
Precipitation: max **36.47in**
Distance: max **441mi**

Dataset 02 · Air Quality in India

Problem Statement

Air pollution in India is responsible for millions of premature deaths each year, yet air quality warnings are often issued reactively rather than in advance. Using daily pollutant readings collected across cities and monitoring stations, **can we predict the air quality category for a given location and day — enabling authorities and citizens to take preventive action before exposure reaches dangerous levels?**

Goal / Aim

Classify the daily air quality level based on pollutant readings and **identify which pollutants are the strongest predictors of poor air quality.**

EDA Findings

Regression or Classification?

Classification — target is **AQI_Bucket** (6 categories: Good, Satisfactory, Moderate, Poor, Very Poor, Severe)

Missing Values

Xylene: **75%** missing.
NH3: **42%**,
PM10: **39%**.
Target column AQI_Bucket: **19%** those rows must be dropped.

Categorical & Continuous Columns

13 Continuous (PM2.5, PM10, NO, NO2, NOx, NH3, CO, SO2, O3, Benzene, Toluene, Xylene, AQI) ·
1 Categorical (Location) ·
1 Datetime (Date)

Outliers

PM2.5 and PM10 both capped at **1000**, AQI max **2049**, O3 max **963**, Benzene max **455**

Dataset 03 · Solar Power Generation

Problem Statement

Inconsistent solar power output creates challenges for grid management, as energy suppliers need to balance supply and demand in real time. Without accurate predictions of how much power a plant will generate, grid operators risk either overloading the network or failing to meet demand. Using inverter-level generation data and plant sensor readings, **can we predict AC power output to support better energy planning and grid stability?**

Goal / Aim

Predict AC power output at a given timestamp using weather sensor readings, and identify which environmental conditions have the strongest influence on power generation.

EDA Findings

Regression or Classification?

Regression — target is **AC_POWER**, a continuous value representing electricity output

Missing Values

Only **4** missing values across the entire dataset, all in the weather columns. Negligible — those rows can be dropped.

Categorical & Continuous Columns

7 Continuous (DC_POWER, AC_POWER, DAILY_YIELD, TOTAL_YIELD …) ·
1 Datetime (DATE_TIME) ·
2 Identifier columns (PLANT_ID, SOURCE_KEY)

Outliers

No significant outliers. Zeros in power and irradiation columns are legitimate nighttime readings, not errors.

Which Dataset to Choose?

	US Accidents	Air Quality India	Solar Power
Size	7.7M rows	137K rows	136K rows
Problem Type	Classification	Classification	Regression
Target Balance	Severe imbalance	Moderate	No imbalance
Missing Values	Moderate	High	Almost none
Predictive Signal	Weak	Strong (0.87)	Very strong (0.88)
Data Coverage	7 years	5 years	34 days
Real World Impact	High	High	Moderate

Air Quality in India has a strong predictive signal, meaningful real world impact, and a well-structured classification problem.